The Complete S_4 Chebotarev–Frobenius Permutation Atlas for the Root Scheme of $x^4 - x - 1$

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Abstract

The integral quartic $f(x) = x^4 - x - 1$ defines an intrinsic prime-indexed family of finite dynamical systems: arithmetic Frobenius $F_p(\alpha) = \alpha^p$ permutes the four geometric roots at every good prime. We prove that $\operatorname{disc}(f) = -283$ and $\operatorname{Gal}(f/\mathbb{Q}) = S_4$, identify factor degrees with primitive cycle lengths, and close the fixed-point and finite-determinant identities on every good fiber. All five S_4 cycle types are exhibited. The final round adds their Chebotarev densities, the non-étale boundary at $p = 283$, and the exact fiberwise Koopman classification. The construction is a source-arithmetic candidate, not a cross-prime target determinant. The universal finite-permutation trace/determinant mechanism remains owned by the prior workspace package HCS-C12A; our owner is the complete $x^4 - x - 1$-specific atlas and its executable convention lock.

1 Quartic arithmetic and the S_4 proof

Put

$$f(x) = x^4 - x - 1, \qquad \mathcal{X} = \operatorname{Spec} \mathbb{Z}[x]/(f). \tag{1}$$

For a rational prime p, write X_p for the geometric roots of the reduction of f in $\overline{\mathbb{F}_p}$. We use *arithmetic Frobenius*

$$F_p : X_p \longrightarrow X_p, \qquad \alpha \longmapsto \alpha^p. \tag{2}$$

Geometric Frobenius is F_p^{-1}. The two conventions have identical cycle lengths, fixed counts, and determinants; this paper never conflates the maps, only their common cycle partition.

Theorem 1 (Arithmetic and Galois closure)**.** *The discriminant of f is -283. The polynomial is irreducible over $\mathbb{Q}$ and its splitting field L has*

$$\operatorname{Gal}(L/\mathbb{Q}) \cong S_4.$$

Every prime $p \neq 283$ is unramified and $\mathcal{X}_{\mathbb{F}_p}$ is a finite étale scheme of degree four. If $\lambda_p = (d_1, \dots, d_k)$ is the multiset of degrees of the distinct irreducible factors of $f \bmod p$, then λ_p is exactly the cycle partition of F_p on X_p.

Proof. For $x^4 + qx + r$ the discriminant is $256r^3 - 27q^4$; inserting $q = r = -1$ gives $-256 - 27 = -283$. Modulo 2,

$$f = x^4 + x + 1.$$

It has no root, and the only monic irreducible quadratic over $\mathbb{F}_2$ is $x^2 + x + 1$, whose square is $x^4 + x^2 + 1$. Hence $f \bmod 2$ is irreducible. Gauss's lemma makes f irreducible over $\mathbb{Q}$, so its Galois group $G \leq S_4$ is transitive. Dedekind's theorem [1, 2] at the unramified prime 2 supplies a 4-cycle. At $p = 7$,

$$f \equiv (x - 3)(x^3 + 3x^2 + 2x - 2) \pmod 7, \tag{3}$$

and the cubic factor has no root, so G also contains a 3-cycle. Consequently 12 divides $|G|$, while $|G|$ divides 24. A subgroup of order 12 in S_4 is the index-two sign kernel A_4, but a 4-cycle is odd; equivalently, the discriminant is not a square. Therefore $G = S_4$.

The discriminant criterion makes precisely $p = 283$ exceptional. At every other prime, the roots are distinct. An irreducible factor of degree d has its roots in one orbit of $\alpha \mapsto \alpha^p$ of length d, and every orbit arises this way [3]. This proves the dictionary. $\square$

Proposition 1 (Five exact witnesses)**.** *Every partition of four occurs among the good fibers. In addition to* (3)*, one has*

$$\begin{aligned} p = 2: &\quad f = x^4 + x + 1 && (4), \\ p = 17: &\quad f = (x+2)(x+5)(x^2 - 7x + 5) && (2+1+1), \\ p = 71: &\quad f = (x^2 + 15x - 20)(x^2 - 15x + 32) && (2+2), \\ p = 83: &\quad f = (x+3)(x+7)(x+14)(x-24) && (1+1+1+1), \end{aligned}$$

where every equality is in the corresponding residue field and the final parenthesis is the Frobenius cycle partition.

Proof. Multiplication gives f modulo the displayed prime. The linear factors are explicit; the remaining quadratic or cubic factors have no residue-field root. The mod-2 case was proved in Theorem 1. ☐

2 All-iterate orbit and determinant closure

HCS-C12A already owns the universal zero-dimensional statement that Frobenius on a reduced finite fiber is a permutation, its iterate counts are traces, and its finite zeta is the reciprocal permutation determinant. C369 does not claim workspace ownership of that universal mechanism. This section specializes it to the five S_4 classes of $x^4 - x - 1$ and locks the factor, fixed, primitive, and determinant conventions in one executable all-good-prime ledger. For a good p, let P_p be the permutation matrix on $\mathbb{C}[X_p]$ defined by $P_p e_\alpha = e_{F_p(\alpha)}$, and set

$$N_p(r) = \#\,\mathrm{Fix}(F_p^r).$$

Theorem 2 (Fixed, primitive, and determinant identities)**.** *Let $p \neq 283$, let $\lambda_p = (d_1, \dots, d_k)$, and let $r, n \geq 1$. Then*

$$N_p(r) = \sum_{j : d_j \mid r} d_j. \tag{4}$$

If $E_p(n)$ is the number of points of exact period n and $C_p(n)$ the number of primitive n-cycles, then

$$E_p(n) = \sum_{d \mid n} \mu(d) N_p(n/d), \qquad C_p(n) = \frac{E_p(n)}{n}. \tag{5}$$

Finally, in $\mathbb{Q}[[u]]$ and therefore as a rational function,

$$\begin{aligned} Z_p(u) &:= \exp\left(\sum_{r \geq 1} \frac{N_p(r)}{r} u^r \right) \\ &= \det(I - uP_p)^{-1} = \prod_{j=1}^{k} (1 - u^{d_j})^{-1}. \end{aligned} \tag{6}$$

Thus the factor degrees are not merely a fixed-point statistic: they are the complete primitive-cycle ledger on the fiber.

Proof. A d-cycle contributes all its d points to $\mathrm{Fix}(F_p^r)$ exactly when $d \mid r$, proving (4). The relation $N_p(r) = \sum_{n \mid r} E_p(n)$ and Möbius inversion give (5); division by n groups exact-period points into cycles. On a d-cycle block C_d, $\det(I - uC_d) = 1 - u^d$. Multiplying blocks proves the determinant equality, while

$$-\log(1 - u^d) = \sum_{m \geq 1} \frac{u^{dm}}{m} = \sum_{m \geq 1} \frac{d}{dm} u^{dm}$$

and (4) prove the exponential equality. ☐

type	$1+1+1+1$	$2+1+1$	$2+2$	$3+1$	4
good primes $\leq 10^4$	43	306	147	411	321
S_4 class size	1	6	3	8	6

Table 1: Exact finite regression counts. The 1,228 entries sum across the row; these counts are not used to prove an asymptotic density.

The canonical evidence enumerates all 1,229 primes at most 10^4, removes the unique bad prime 283, and checks (4)–(6) for $1 \le r \le 12$ on all 1,228 good fibers: 14,736 prime–iterate cells. A separate factorization backend recomputes every partition. Exact symbolic and hostile-mutation lanes test the proof anchors and schema. Finite enumeration is an implementation receipt; Theorems 1 and 2 establish the quartic specialization for all good primes and all iterates.

3 Chebotarev density, bad fiber, and operator boundary

This round completes the quartic-specific density, bad-fiber, and route boundary.

Theorem 3 (Complete density and boundary atlas)**.** *The natural densities of good primes with the five factor partitions are*

λ_p	$1+1+1+1$	$2+1+1$	$2+2$	$3+1$	4
density	1/24	1/4	1/8	1/3	1/4

At the excluded prime,

$$f \equiv (x-115)(x-93)^2(x+18) \pmod{283}, \qquad \gcd(f, f') = x - 93. \tag{7}$$

Hence $\mathcal{X}_{\mathbb{F}_{283}}$ is non-étale; no four-distinct-point permutation atlas is asserted there.

Proof. The conjugacy classes of S_4 with cycle types $1{+}1{+}1{+}1, 2{+}1{+}1, 2{+}2, 3{+}1, 4$ have sizes $1, 6, 3, 8, 6$. By Theorem 1, Dedekind's dictionary identifies these classes with the factor partitions. Chebotarev's density theorem divides each class size by $|S_4| = 24$ [2]. Multiplication verifies (7); the common root 93 of f and f' proves nonreducedness. ☐

Proposition 2 (Natural fiber quantization)**.** *For each good p, P_p is a canonical unitary on the four-dimensional Hilbert space $\ell^2(X_p)$. It is self-adjoint exactly for types $1+1+1+1$, $2+1+1$, and $2+2$. For types $3+1$ and 4 it remains unitary but is not self-adjoint.*

Proof. P_p permutes an orthonormal basis, so $P_p^* = P_p^{-1}$. It is self-adjoint precisely when $P_p = P_p^{-1}$, or $F_p^2 = I$; this is equivalent to every cycle having length at most two. ☐

4 Route decision and nonclaims

The integral scheme supplies prime fibers, prime-power Frobenius iterates, Chebotarev classes, and source-derived primitive weights. It therefore earns $A0_{\text{STRUCTURAL\_ARITHMETIC\_RELATION}}$. The exact rational finite-fiber determinant in (6) earns the deliberately local $A1_{\text{PASS\_ANALYTIC}}$ as an exact applicability result, not as new ownership of the universal mechanism. Proposition 2 gives $A4_{\text{NATURAL\_QUANTIZATION}}$.

The prime p indexes a different fiber; it is not a time step of one autonomous system. We construct no cross-prime Fredholm direct sum, no determinant-class regularization, and no target Euler product. There is no target continuation, functional equation, divisor, zero fit, or counting law. Thus $A2_{\text{FAIL}}$ and $A3_{\text{FAIL}}$. The overall verdict is `ROUTE_A_ARITHMETIC_CANDIDATE`, under the literal scope `NO_BAD_EULER_OR_ROOT_NUMBER`. Fiber permutation unitaries are not a Hilbert–Pólya operator, and Route B is not invoked.

References

[1] J.-P. Serre, *Topics in Galois Theory*, second edition, A K Peters, 2008.

[2] J. Neukirch, *Algebraic Number Theory*, Springer, 1999.

[3] R. Lidl and H. Niederreiter, *Finite Fields*, second edition, Cambridge University Press, 1997.